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IUT STUDENT BRANCH CHAPTER

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# PCB DESIGN CONTEST

## ROUND TWO

# General Rules

1. Participants must design the PCB according to the given schematic, design requirements, or component list provided in the contest question.
2. Any form of plagiarism or use of pre-made full designs (other than reference libraries) will result in immediate disqualification.
3. The design must be created using the specified EDA tool (**EasyEDA**) as mentioned in the question.
4. **File Submission Requirements:**

   Participants must submit the following files **individually** (*ZIP or compressed file formats are strictly prohibited*):

   - **Main project files:** Both **Schematic** and **Board** files.
   - **PDF layer files:** A **single PDF** containing **all layers** of the PCB. Each page must be **clearly labeled** with the corresponding layer name (e.g. Top Layer, Bottom Layer, Silkscreen, Drill Map etc.).
   - **Report:** A detailed **PCB Design Report** including 2D and 3D views of the final PCB layout.
   - **File naming:** name_uni.filetype
     i.e. sadi_iut.sch/.pdf/.json

All required files must be submitted **before the announced deadline**

Late submissions will not be evaluated.

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# Evaluation Criteria

| Category | Description | Marks |
|---|---|---|
| **1. Schematic Design** | Proper circuit connections, component labeling, net naming, and clarity of design. | 15-20% |
| **2. Custom Library Creation** | For creating custom symbols or footprints. Follow given specifications in the question (if any) or datasheet. | 5–10% |
| **3. PCB Layout Design** | Component placement, routing optimization, trace width selection, power/ground planes, and design compactness. | 25-30% |
| **4. Design Rule Check (DRC) Compliance** | Must pass all DRC checks. For every logical DRC error, **5 marks** will be deducted. | 15-20% |
| **5. Creativity and Design Efficiency** | Compact and efficient routing, smart component placement, and thermal considerations. | 10-15% |
| **6. Aesthetics and Documentation** | Silkscreen clarity, layer management, alignment, proper text placement, and overall presentation. | 10-15% |

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# Additional Notes

- You may need to build libraries according to the given details; if not specified, follow the component's **official datasheet**.
- Designs with **missing connections**, **incorrect net assignments**, or **footprint mismatches** will face point deductions.
- Bonus marks (up to **5 Marks**) may be awarded for **innovative routing** or **excellent PCB design aesthetics**.
- **Alternative versions** of any components, whether through-hole or SMD, may be used in the design if they **meet the given specifications.**

# Problem 1: SparkFun OpenLog Artemis Data Logger (110 points)

Design a compact embedded data-logging PCB based on the SparkFun OpenLog Artemis architecture. The board should be capable of acquiring motion and sensor data, storing the collected data on a microSD card, and providing USB, serial, and QWIIC/I²C connectivity for external sensors and configuration. The design should integrate the Artemis processor section, IMU/sensor circuitry, microSD interface, USB interface, LiPo charging and power-management circuitry, external sensor connectors, and status indication.

**Type:** Precision Analog / Data Acquisition **Design**

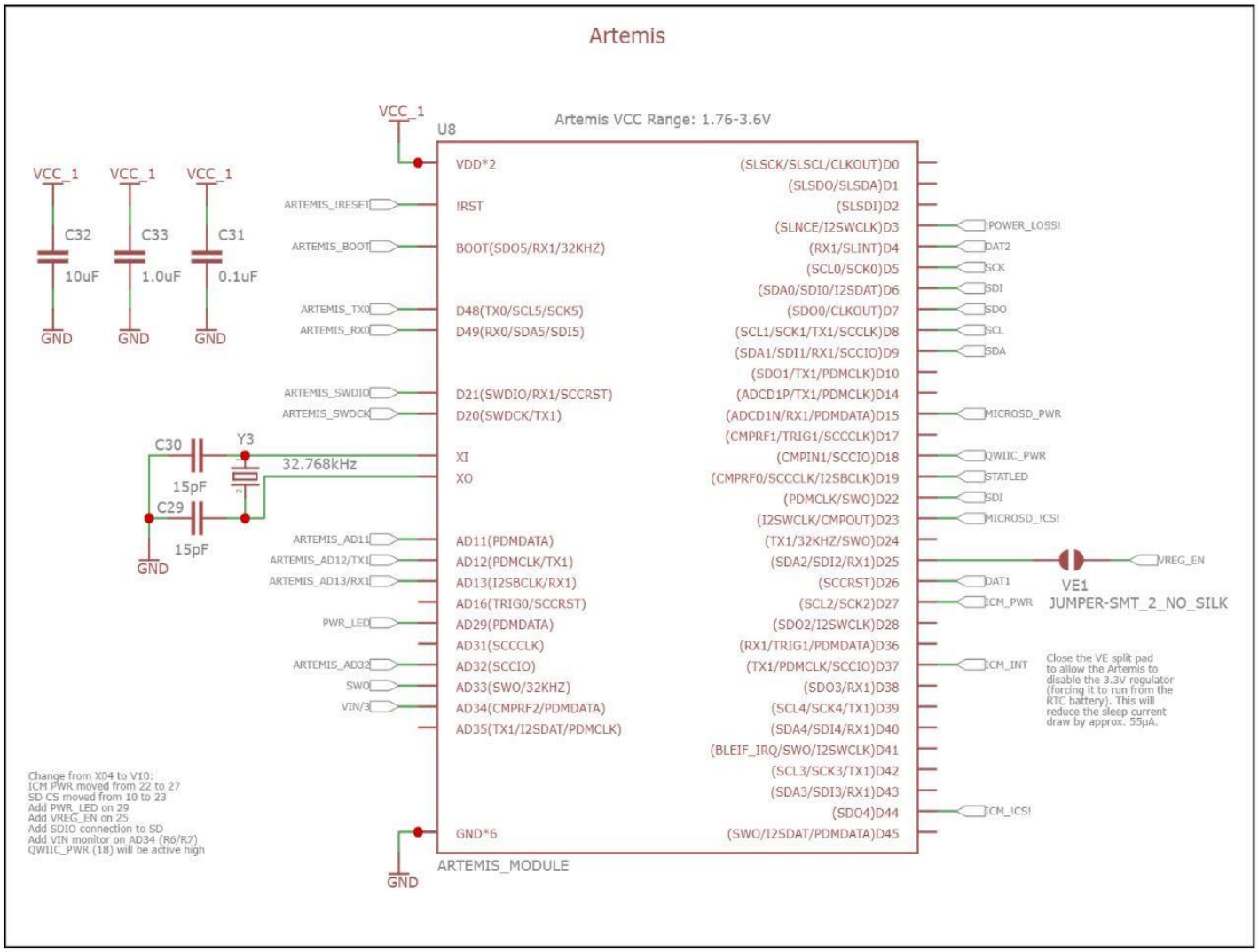
## Requirements:

- Use the SparkFun OpenLog Artemis schematic as the reference for the circuit design.
- Place the Artemis module/MCU and its decoupling components close together.
- Keep the IMU and other sensitive sensor circuitry away from noisy power and USB sections.
- Place the microSD connector near the PCB edge for easy user access.
- Keep SPI traces between the MCU and microSD card short and organized.
- Place USB connector and USB protection/interface components near the board edge.
- Keep USB D+ and D− routing short, direct, and reasonably matched.
- Place the LiPo connector, charger IC, and associated power components together as a dedicated power section.
- Keep high-current charging paths away from sensor signal traces.
- Place Qwiic/I²C connectors along the PCB perimeter and keep SDA/SCL routing short.
- Provide accessible programming/debug and serial interface connections.
- Use proper ground-plane coverage, especially beneath the MCU, sensor, and digital sections.
- Place decoupling capacitors as close as possible to the corresponding IC supply pins.
- Clearly mark connector names, polarity, pin functions, and important interfaces on the silkscreen.
- The PCB should be compact, neatly routed, and pass all applicable DRC checks.
- Participants may add useful test points, mounting holes, or labels if required.

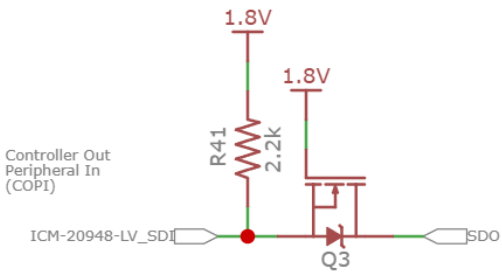
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## The Schematic:

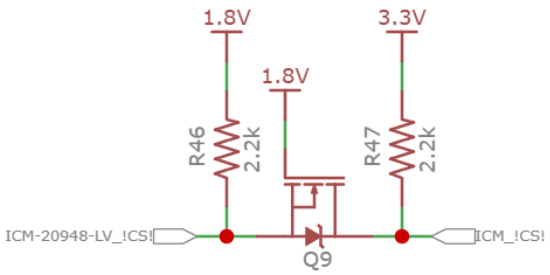
Follow this schematic for designing the PCB.



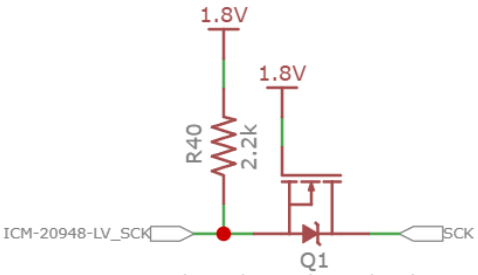
## Level Shifting



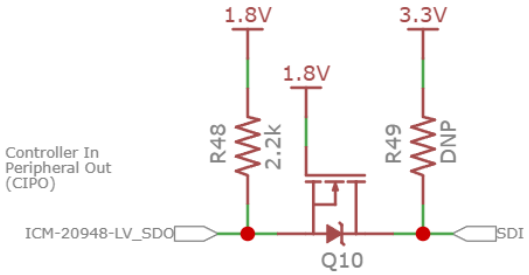

20V/0.2A/8MHz/1.2Ω/1Vth




20V/0.2A/8MHz/1.2Ω/1Vth

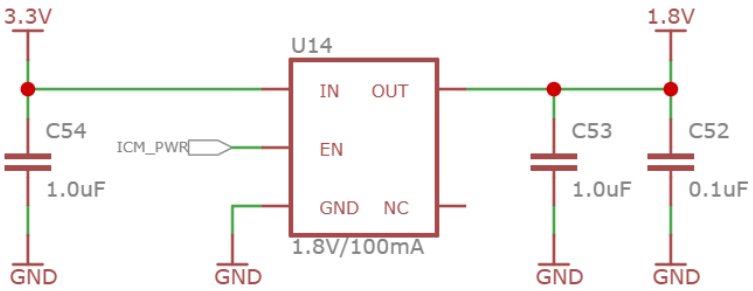


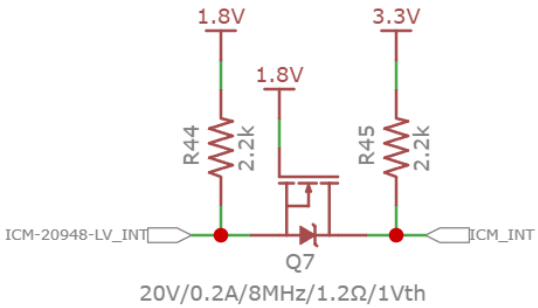

20V/0.2A/8MHz/1.2Ω/1Vth




20V/0.2A/8MHz/1.2Ω/1Vth

## Voltage Regulator

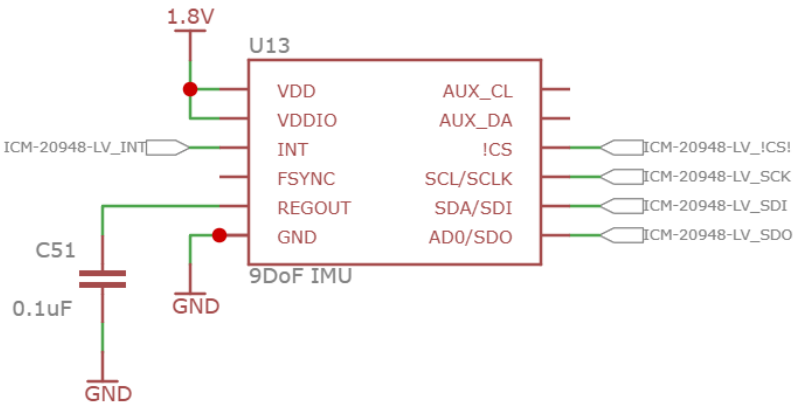



20V/0.2A/8MHz/1.2Ω/1Vth

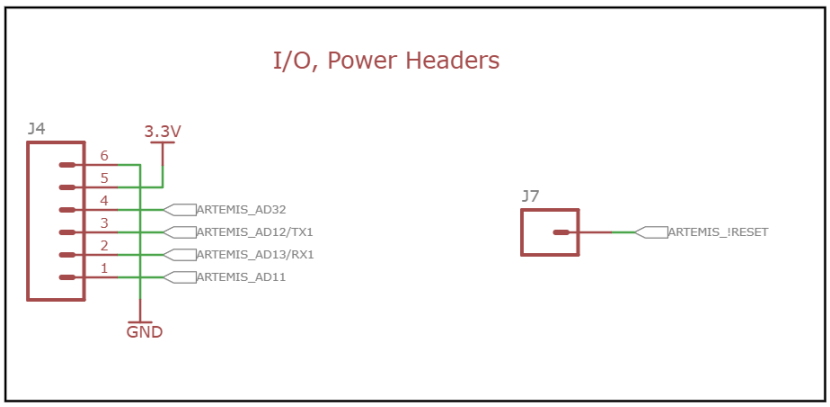
SP6214-1.8V
---
Iout (max): 100mA
Vin (max): 7.0V
Vdrop (max): 250mV
Iq: 65uA

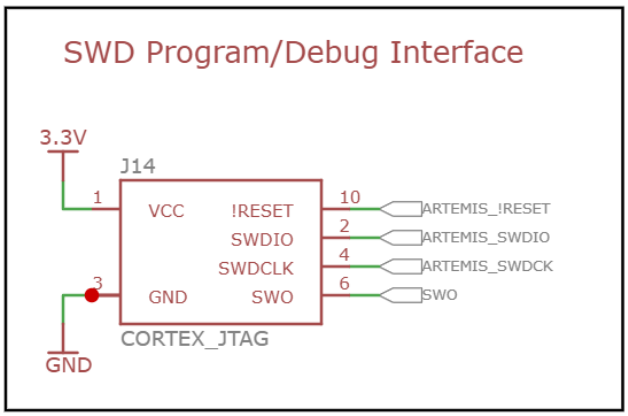
## ICM-20948 IMU

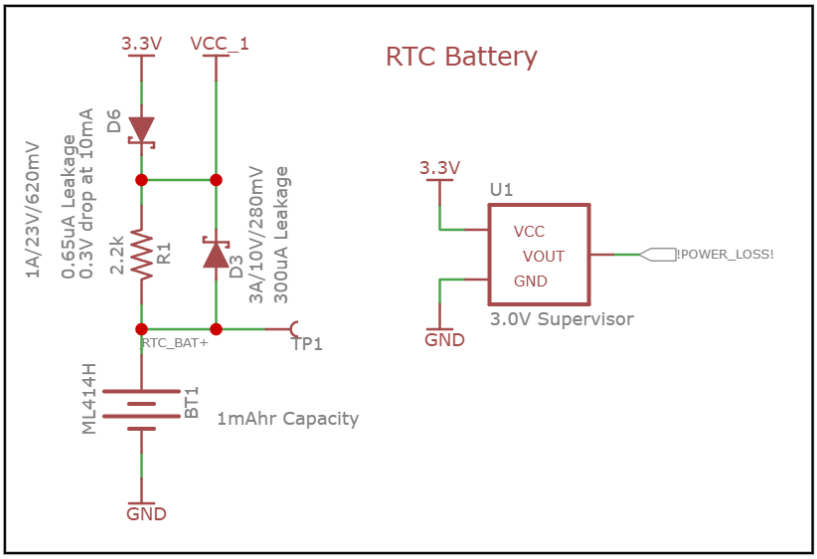


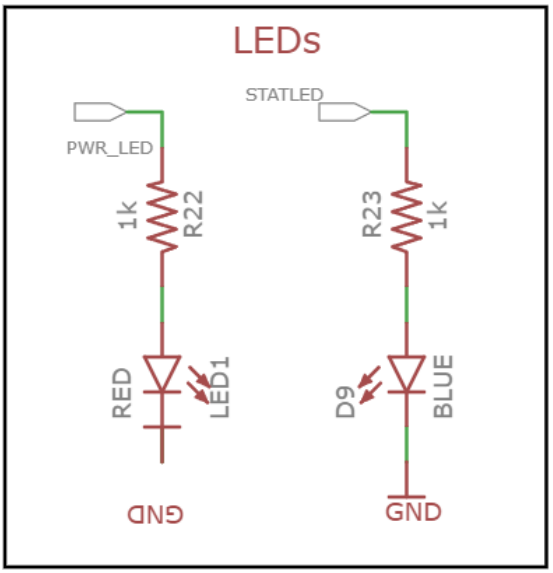
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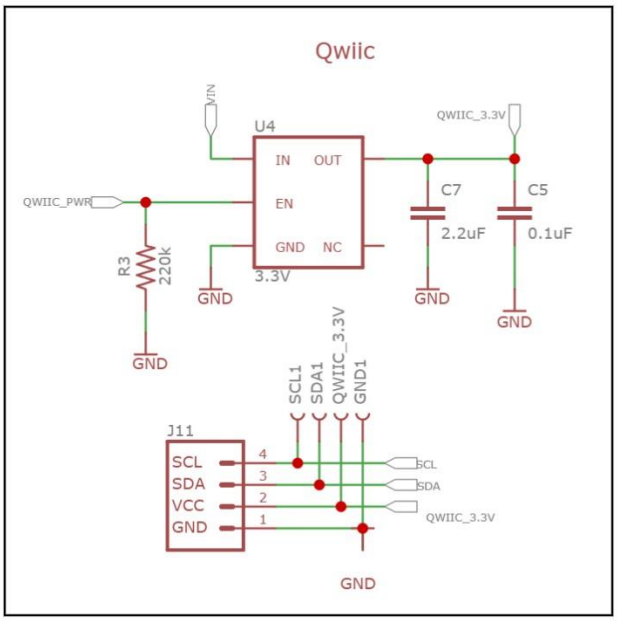
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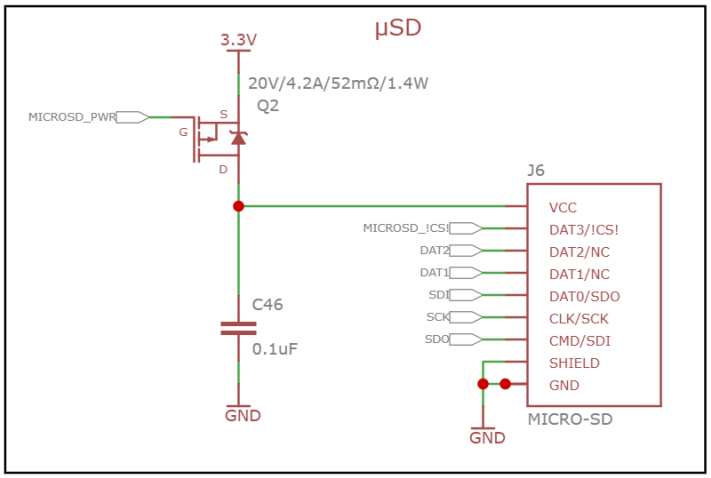


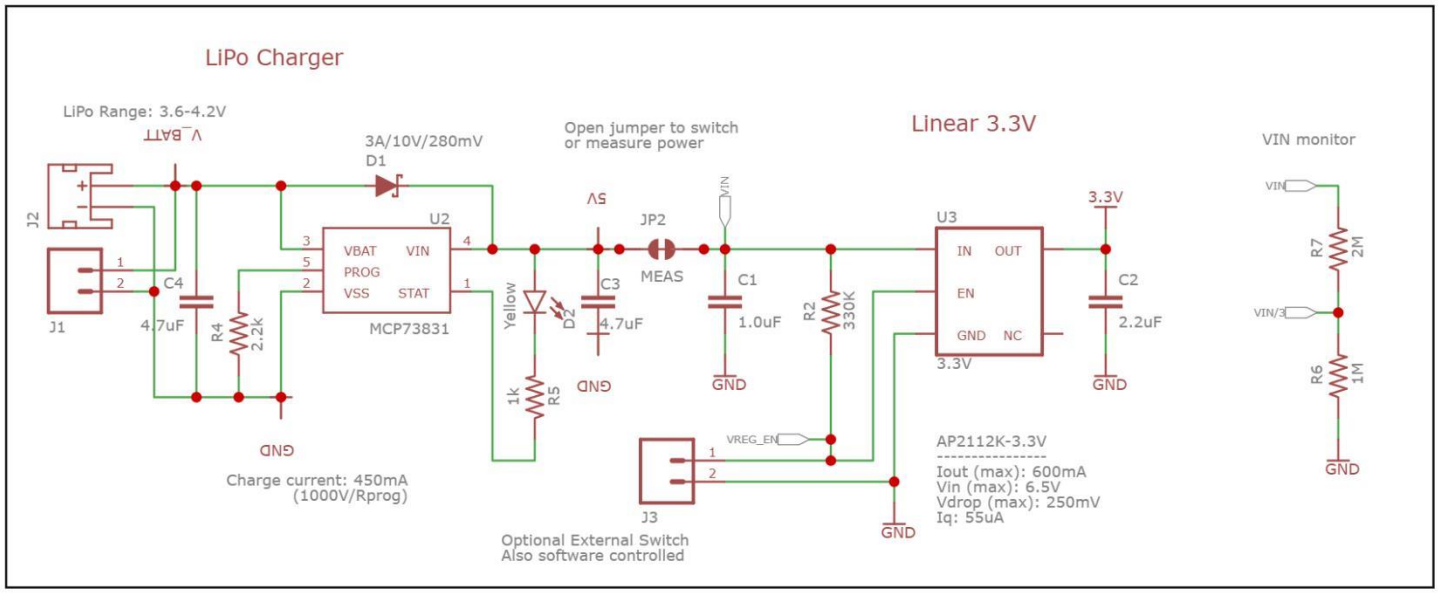



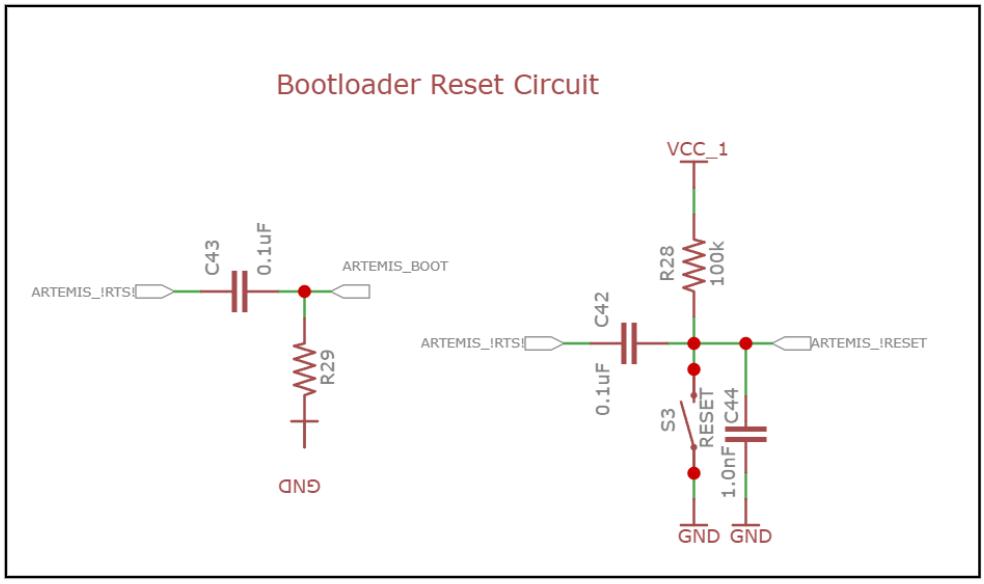



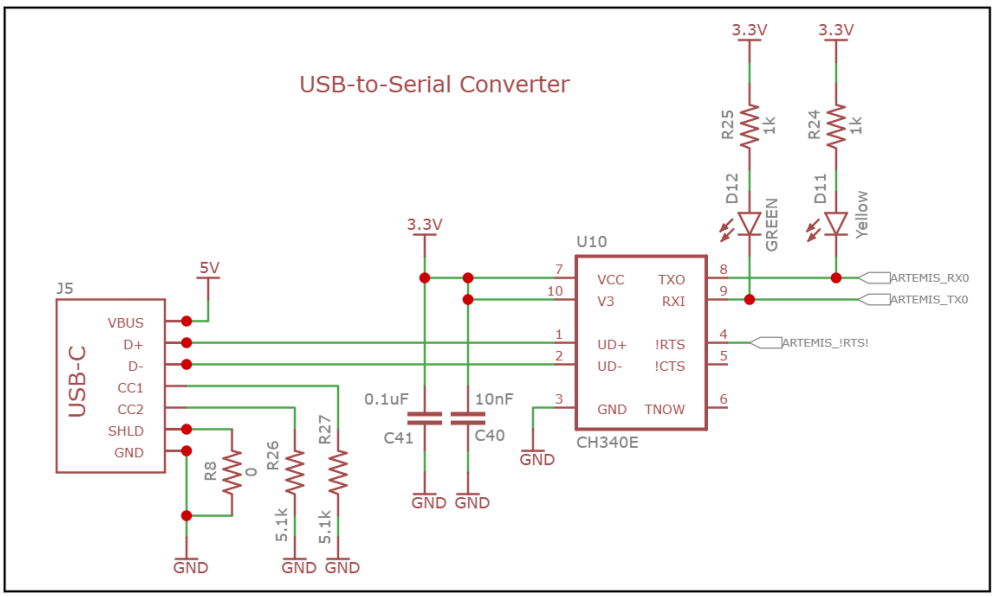




LiPo Charger
LiPo Range: 3.6-4.2V
V_BATT
3A/10V/280mV
D1
Open jumper to switch
or measure power
Linear 3.3V
VIN monitor
J2
J1
C4
4.7uF
R4
2.2k
U2
VBAT
PROG
VSS
VIN
STAT
MCP73831
5V
JP2
MEAS
VIN
C1
1.0uF
R2
330K
U3
IN
OUT
EN
GND
NC
3.3V
C2
2.2uF
Yellow
D2
C3
4.7uF
1k
R5
GND
VREG_EN
J3
Optional External Switch
Also software controlled
Charge current: 450mA
(1000V/Rprog)
AP2112K-3.3V
Iout (max): 600mA
Vin (max): 6.5V
Vdrop (max): 250mV
Iq: 55uA
R7
2M
R6
1M
VIN/3


Bootloader Reset Circuit
C43
0.1uF
ARTEMIS_!RTS!
ARTEMIS_BOOT
R29
GND
VCC_1
R28
100k
C42
0.1uF
ARTEMIS_!RESET
S3
RESET
C44
1.0nF
GND


USB-to-Serial Converter
3.3V
3.3V
R25
1k
R24
1k
D12
GREEN
D11
Yellow
J5
USB-C
VBUS
D+
D-
CC1
CC2
SHLD
GND
5V
U10
VCC
V3
TXO
RXI
UD+
UD-
!RTS
!CTS
GND
TNOW
CH340E
ARTEMIS_RX0
ARTEMIS_TX0
ARTEMIS_!RTS!
0.1uF
C41
10nF
C40
R8
0
R26
5.1k
R27
5.1k

## Component List:

| No. | Component | Designation |
|---|---|---|
| 1. | Artemis Module (ARTEMIS_MODULE) | U8 |
| 2. | 9DoF IMU ICM-20948 | U13 |
| 3. | USB-to-Serial Converter CH340E | U10 |
| 4. | LiPo Charger MCP73831 | U2 |
| 5. | 3.3V LDO Regulator AP2112K-3.3V | U3, U4 |
| 6. | 1.8V LDO Regulator SP6214-1.8V(an equivalent 1.8V LDO may be used) | U14 |
| 7. | 3.0V Supervisor | U1 |
| 8. | MOSFET 20V/4.2A/52mΩ/1.4W (microSD power switch) | Q2 |
| 9. | MOSFET 20V/0.2A/8MHz/1.2Ω/1Vth (level shifting) | Q1, Q3, Q7, Q9, Q10 |
| 10. | Schottky Diode 3A/10V/280mV | D1, D3 |
| 11. | Diode 1A/23V/620mV | D6 |
| | | |
| 13. | Resistor (R49 = DNP) | R1–R8, R22–R29, R40, R41, R44–R49 |
| 14. | Capacitor | C1–C5, C7, C29–C33, C40–C44, C46, C51–C54 |
| 15. | Crystal 32.768kHz | Y3 |
| 16. | USB Type-C Connector | J5 |
| 17. | microSD Socket | J6 |
| 18. | Qwiic Connector (JST 4-pin, 1mm) | J11 |
| 19. | LiPo Battery Connector (JST 2mm) | J2 |
| 20. | Cortex JTAG/SWD Header (2x5, 1.27mm) | J14 |
| 21. | Header Pins (1x2, 1x6, 1x1) | J1, J3, J4, J7 |
| 22. | Reset Tactile Switch | S3 |
| 23. | RTC Battery ML414H | BT1 |
| 24. | Jumpers (MEAS, SMT jumper) | JP2, VE |
| 25. | Test Points | GND, QWIIC_3.3V, SCL, SDA, TP1 |

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## Datasheets:

- ICM-20948.pdf
- CH340E
- MCP73831.pdf
- AP2112K-3.3.pdf

## Note:

If your designing software doesn't have any footprint of any particular component, you may make or download a suitable footprint from the internet. In that case, follow the dimensions given in the datasheets.

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# Problem 2: Infineon REF-FRIDGE-C101T-6ED Motor/Inverter Controller (70 points)

Design a three-phase inverter/control PCB suitable for driving a BLDC/PMSM motor, based on the REF-FRIDGE-C101T-6ED reference architecture. The design should integrate the gate-drive circuitry, power-switching section, sensing, and control electronics while maintaining safe separation between high-current switching and low-level control circuitry.

**Type:** Motor Control / Power Electronics

**Editable source:** Altium Designer

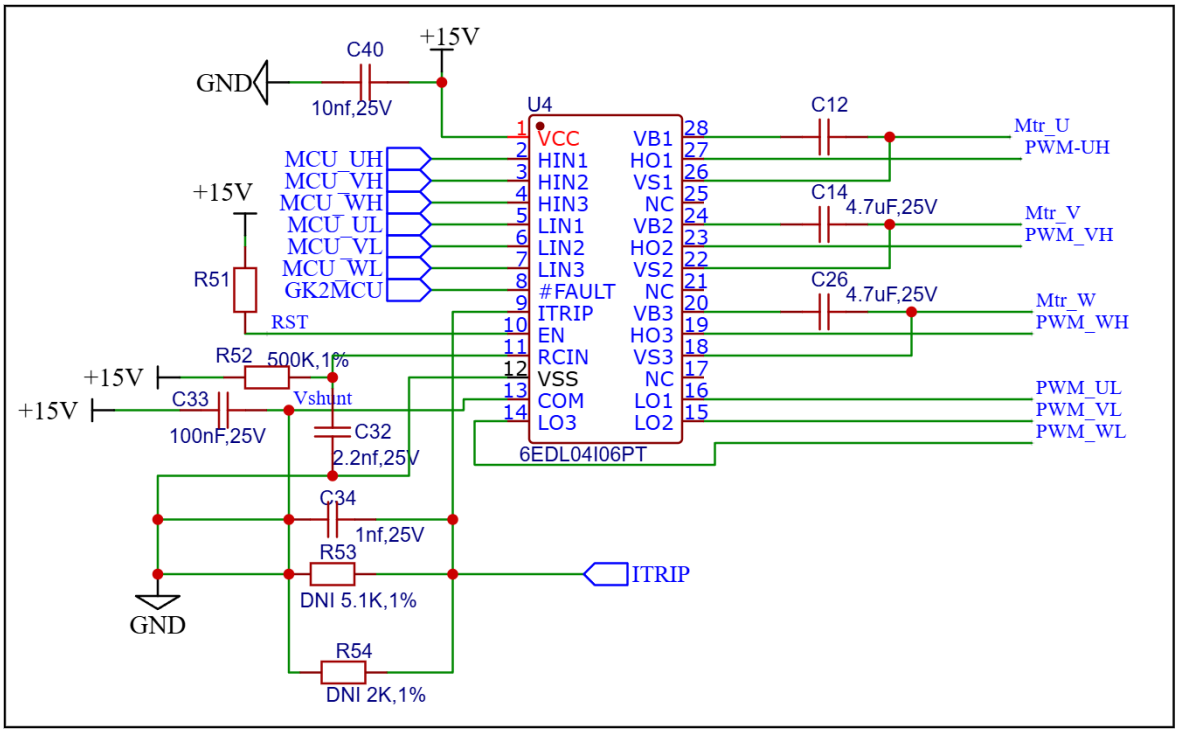
## Design Requirements:

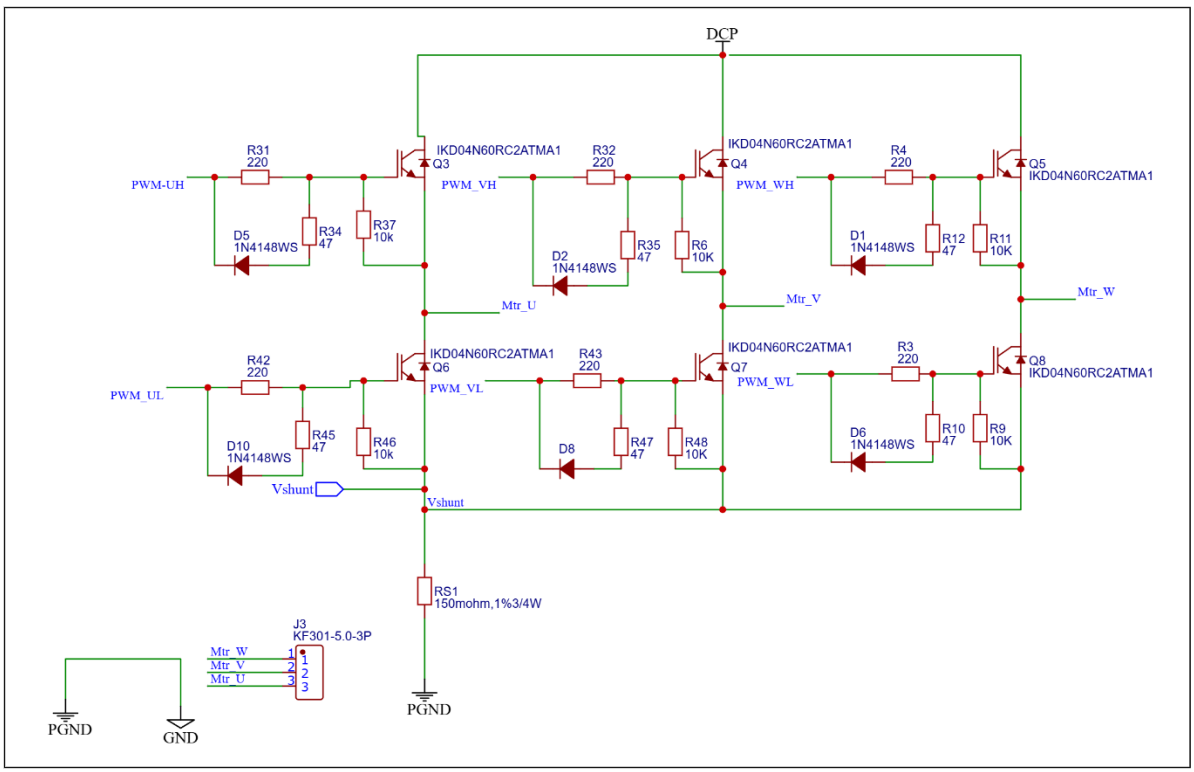
- Arrange the three phase/power stages in a logical and compact power flow.
- Keep gate-drive connections short and direct.
- Minimize high-current DC-link loop area.
- Keep switching nodes away from sensitive sensing and control signals.
- Use appropriately sized copper areas/traces for the current-carrying paths.
- Provide thermal copper and via structures around power components where appropriate.
- Maintain suitable creepage and clearance between high-voltage and lowvoltage/control areas.
- Provide useful test points for DC bus, phase, drive, and control supply signals.

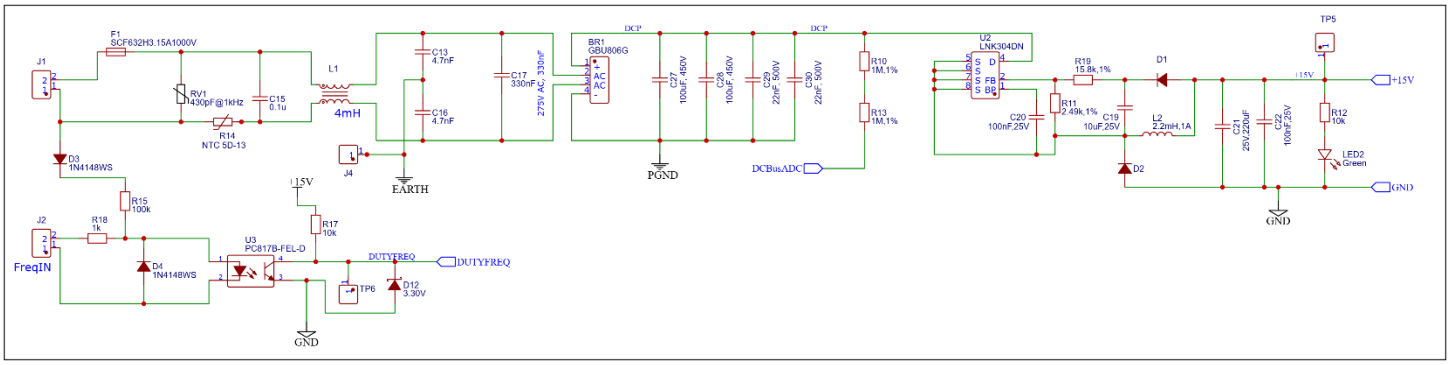
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## The Schematic:

Follow this schematic for designing the PCB.






# Component List:

| No. | Component | Designation |
|---|---|---|
| 1. | Off-line Switcher LNK3204D | U2 |
| 2. | Optocoupler PC817 | U3 |
| 3. | 3-Phase Gate Driver 6EDL04I06PT | U4 |
| 4. | IGBT IKD04N60RC2 | Q3–Q8 |
| 5. | Bridge Rectifier GBU806 | BR |
| 6. | Fuse T3.15A 250V | F1 |
| 7. | Varistor 300Vrms / 385Vdc | RV1 |
| 8. | NTC Thermistor 5D-13 | R14 |
| 9. | Common-Mode Choke 4mH | L1 |
| 10. | Shunt Resistor 150mΩ, 1%, 3/4W | RS1 |
| 11. | Connectors (CON-2PIN, FreqIN, CON-3PIN, TJC2-1) | J1, J2, J3, J4 |
| 12. | Test Points | TP5, TP6 |
| 13. | Diodes, LEDs, Resistors, Capacitors, Inductor | As shown in the schematic |

## Datasheets:

- LNK3204D.pdf
- 6EDL04I06PT.pdf
- IKD04N60RC2.pdf
- PC817.pdf
- GBU806.pdf

## Note:

If your designing software doesn't have any footprint of any particular component, you may make or download a suitable footprint from the internet. In that case, follow the dimensions given in the datasheets.

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# Problem 3: STM32 Ethernet + CAN + RS-485 Industrial Gateway (120 points)

Design a compact industrial communication gateway that bridges Ethernet, CAN, and RS-485 for factory or robotics applications.

**Type:** High-Speed Digital / USB / USB-C

**Editable source:** Altium Designer

## Design Requirements:

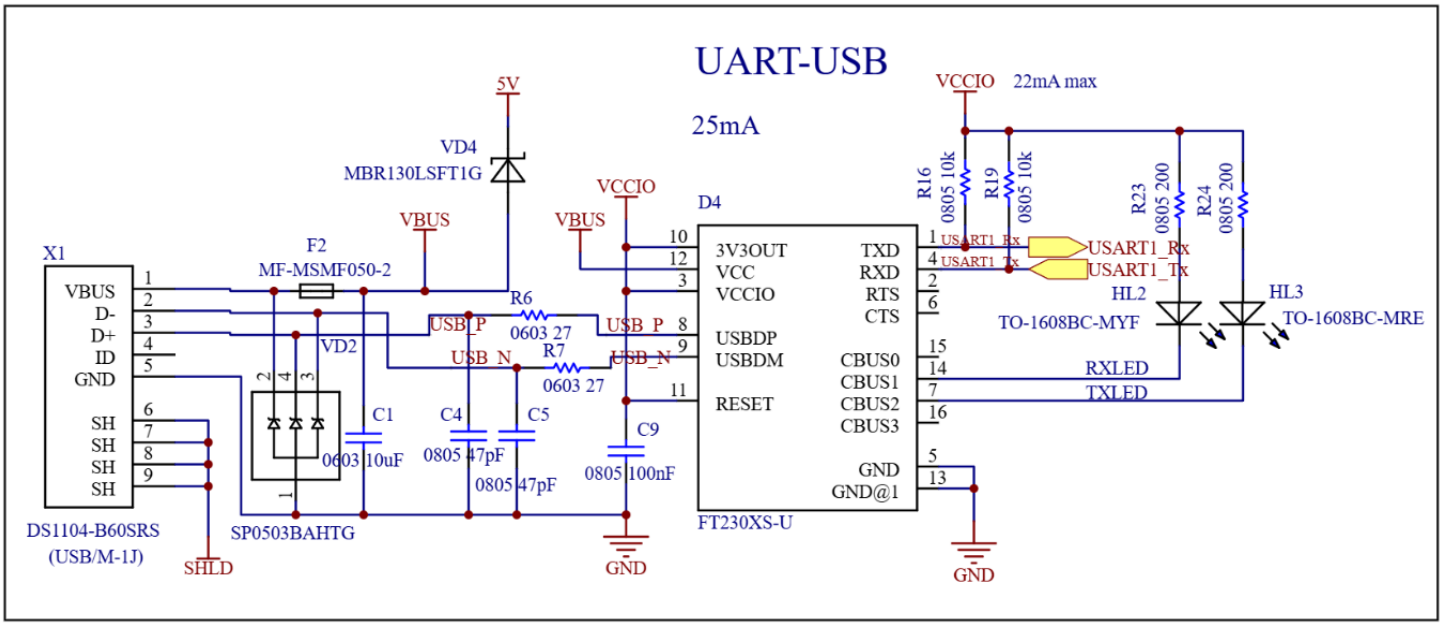
- Ethernet TX/RX differential routing
- PHY-to-RJ45 placement
- CAN and RS-485 protection
- Termination resistor placement
- Isolation/ground partitioning
- 24 V power input and buck stage
- MCU decoupling
- Connector placement
- EMI-aware routing
- Test points and clean silkscreen

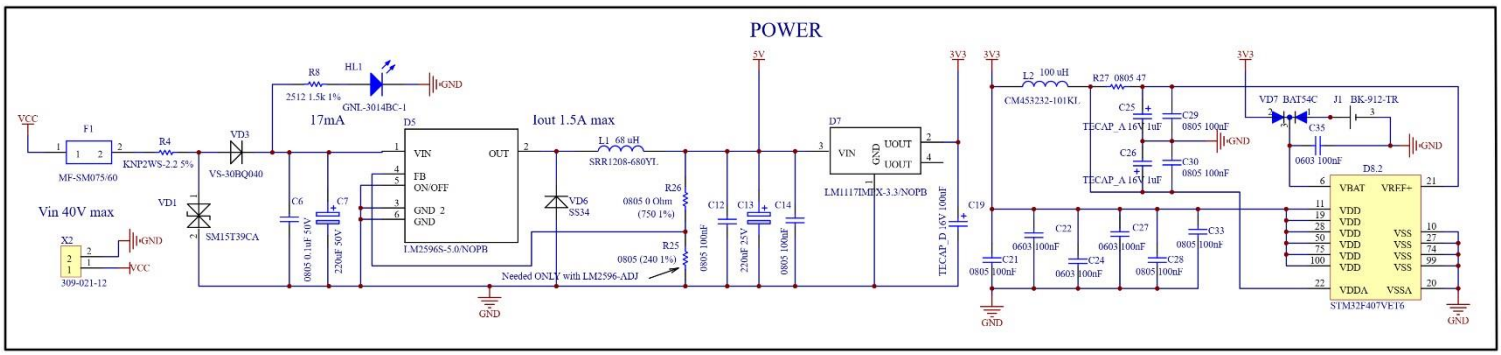
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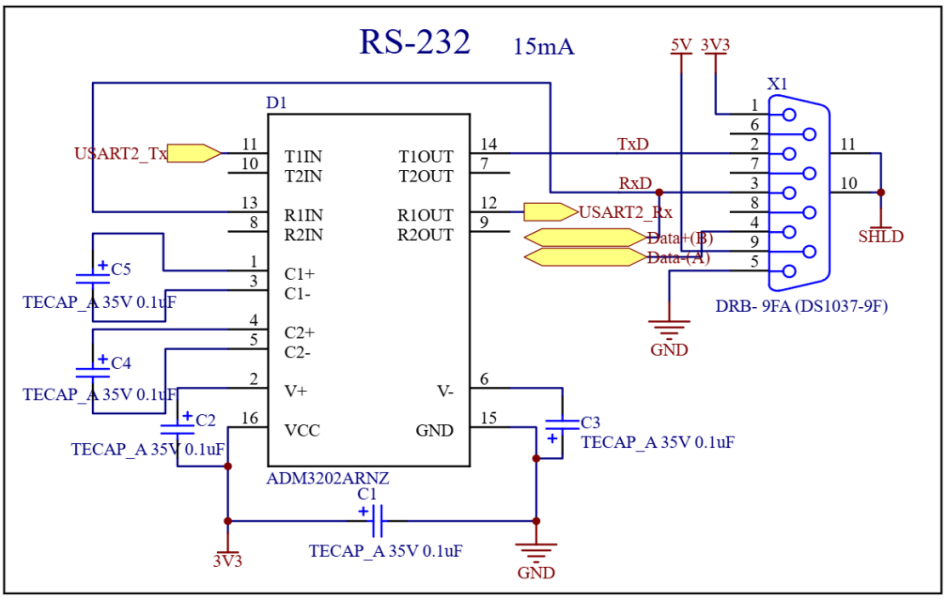
Follow this schematic for designing the PCB.

## The Schematic:

Follow this schematic for designing the PCB.

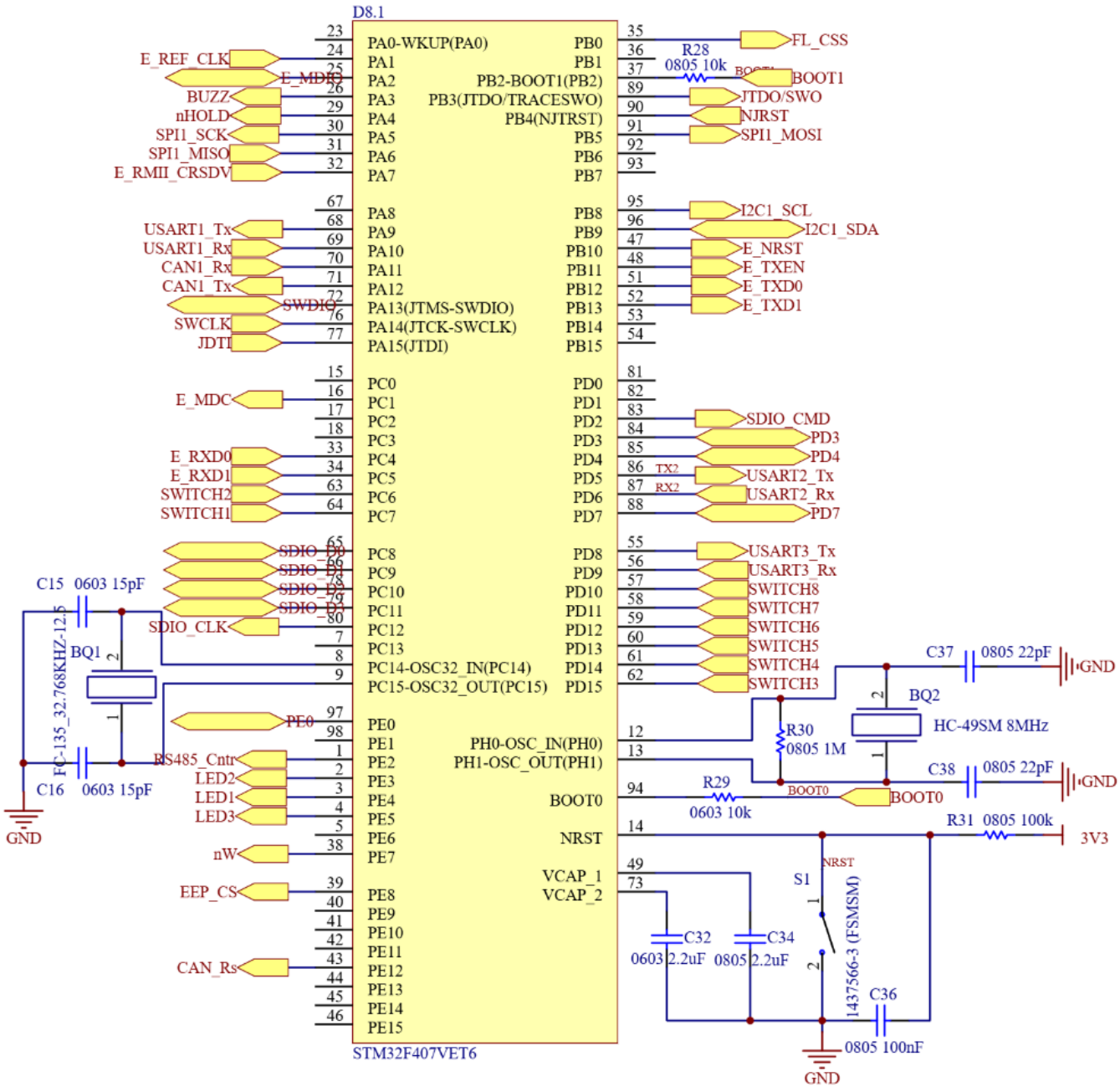


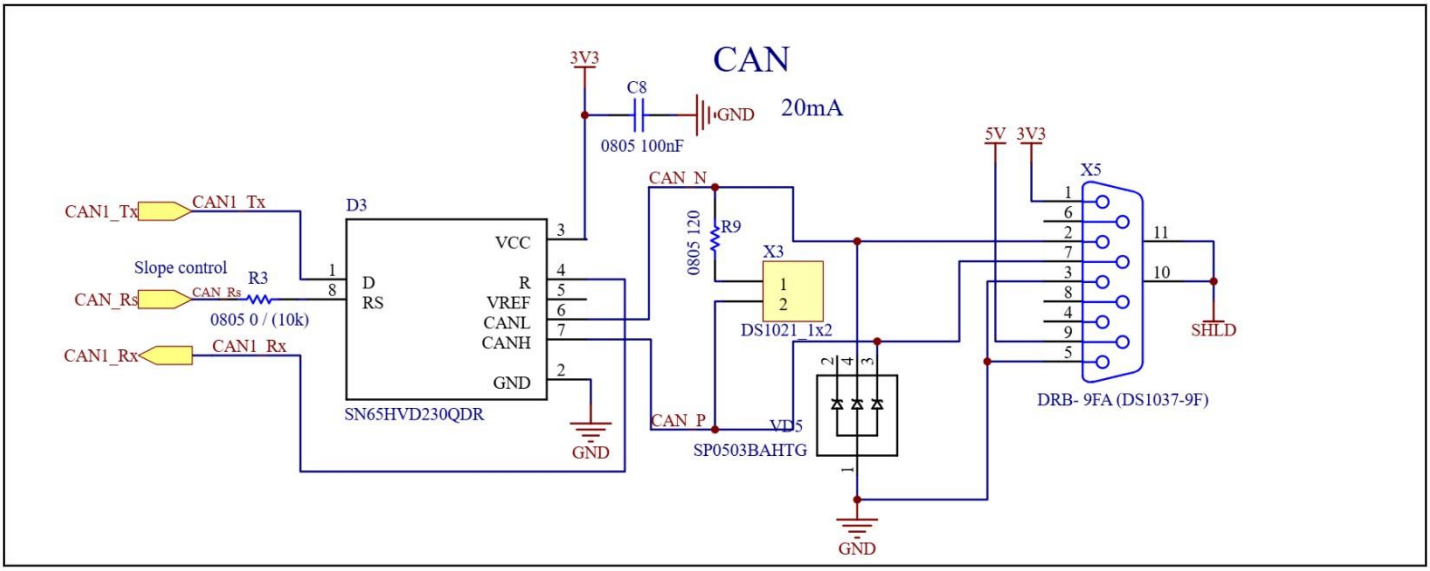



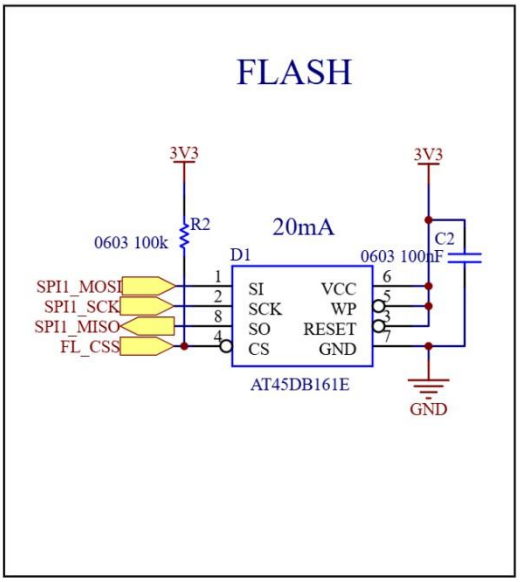



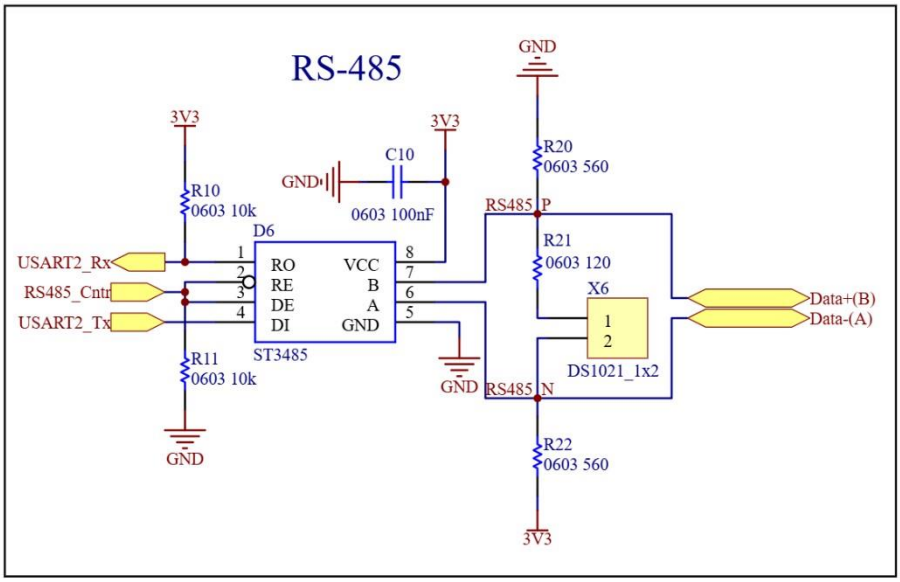
# MCU

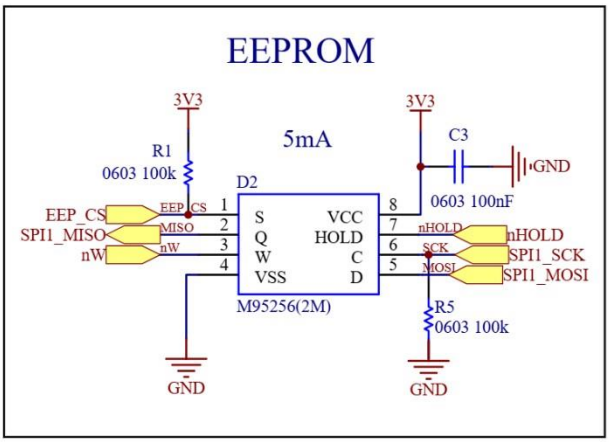
240mA

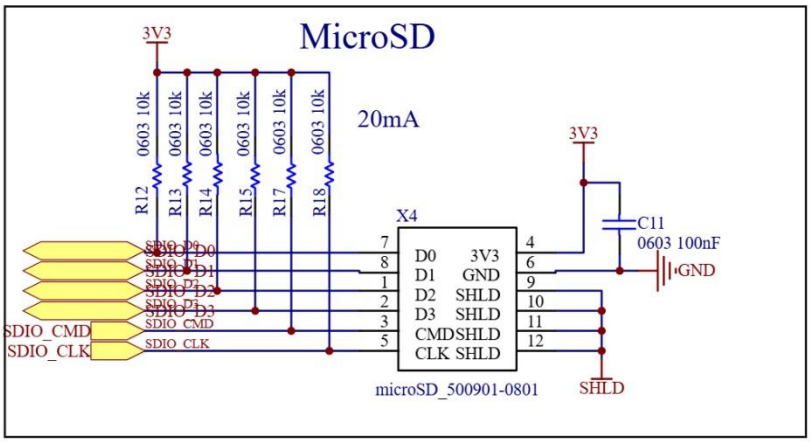

D8.1
STM32F407VET6
PA0-WKUP(PA0)
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13(JTMS-SWDIO)
PA14(JTCK-SWCLK)
PA15(JTDI)
PB0
PB1
PB2-BOOT1(PB2)
PB3(JTDO/TRACESWO)
PB4(NJTRST)
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14-OSC32_IN(PC14)
PC15-OSC32_OUT(PC15)
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD7
PD8
PD9
PD10
PD11
PD12
PD13
PD14
PD15
PE0
PE1
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PE14
PE15
PH0-OSC_IN(PH0)
PH1-OSC_OUT(PH1)
BOOT0
NRST
VCAP_1
VCAP_2
E_REF_CLK
E_MDIO
BUZZ
nHOLD
SPI1_SCK
SPI1_MISO
E_RMII_CRSDV
USART1_Tx
USART1_Rx
CAN1_Rx
CAN1_Tx
SWDIO
SWCLK
JDTI
E_MDC
E_RXD0
E_RXD1
SWITCH2
SWITCH1
SDIO_D0
SDIO_D1
SDIO_D2
SDIO_D3
SDIO_CLK
RS485_Cntr
LED2
LED1
LED3
nW
EEP_CS
CAN_Rs
FL_CSS
R28
0805 10k
BOOT1
JTDO/SWO
NJRST
SPI1_MOSI
I2C1_SCL
I2C1_SDA
E_NRST
E_TXEN
E_TXD0
E_TXD1
SDIO_CMD
PD3
PD4
TX2
USART2_Tx
RX2
USART2_Rx
PD7
USART3_Tx
USART3_Rx
SWITCH8
SWITCH7
SWITCH6
SWITCH5
SWITCH4
SWITCH3
C15 0603 15pF
BQ1
FC-135_32.768KHZ-12.5
C16 0603 15pF
GND
C37 0805 22pF
BQ2
HC-49SM 8MHz
R30
0805 1M
C38 0805 22pF
R29
0603 10k
BOOT0
R31 0805 100k
3V3
S1
NRST
1437566-3 (FSMSM)
C32
0603 2.2uF
C34
0805 2.2uF
C36
0805 100nF

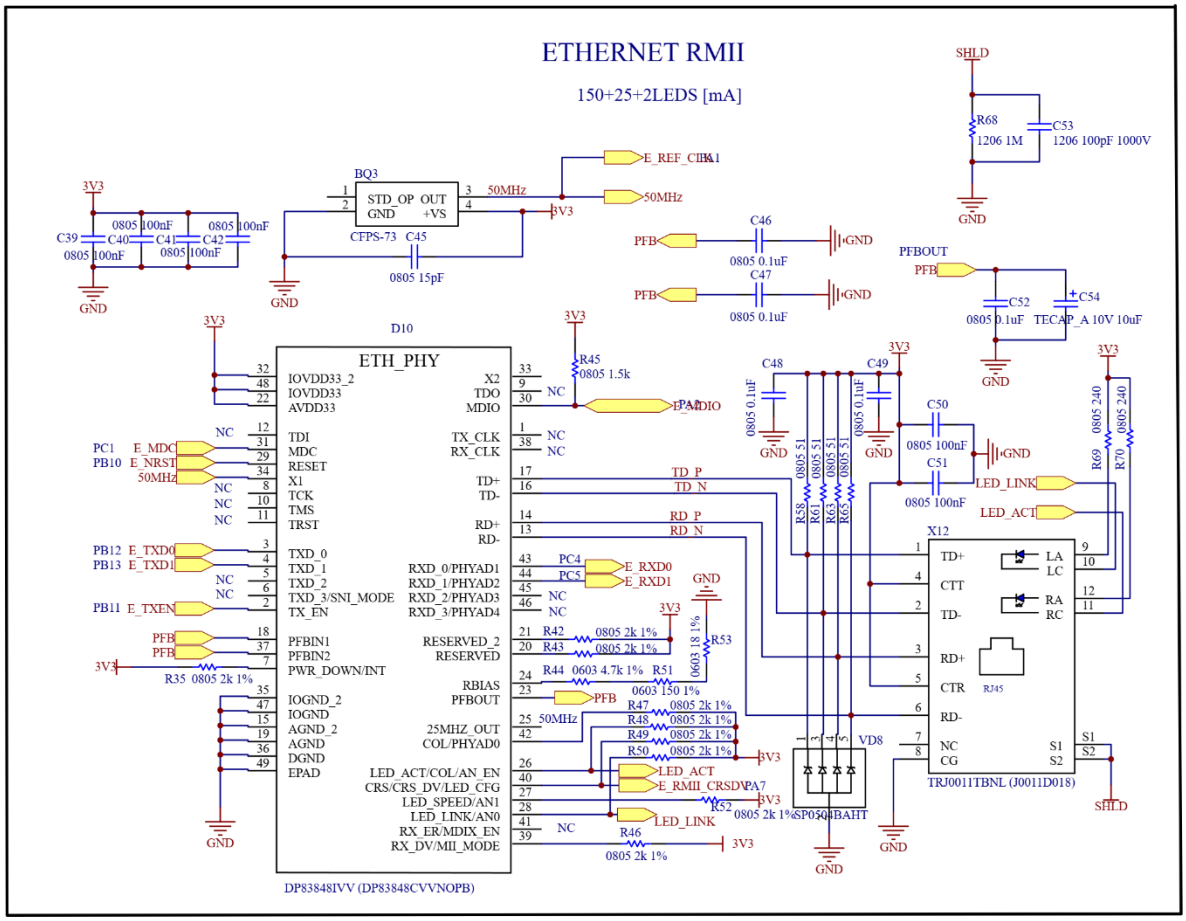

CAN
3V3
C8
0805 100nF
GND
20mA
5V
3V3
X5
CAN_N
CAN1_Tx
CAN1_Tx
D3
VCC
R9
0805 120
X3
Slope control
CAN_Rs
CAN_Rs
R3
0805 0 / (10k)
D
RS
R
VREF
CANL
CANH
GND
DS1021_1x2
SHLD
CAN1_Rx
CAN1_Rx
SN65HVD230QDR
CAN_P
VD5
SP0503BAHTG
DRB- 9FA (DS1037-9F)
GND
GND

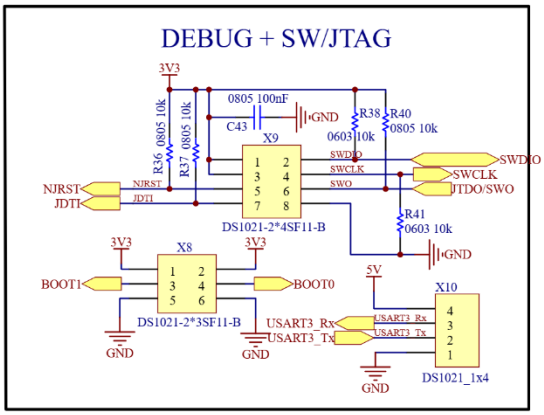

FLASH
3V3
3V3
R2
0603 100k
20mA
D1
C2
0603 100nF
SPI1_MOSI
SPI1_SCK
SPI1_MISO
FL_CSS
SI
SCK
SO
CS
VCC
WP
RESET
GND
AT45DB161E
GND

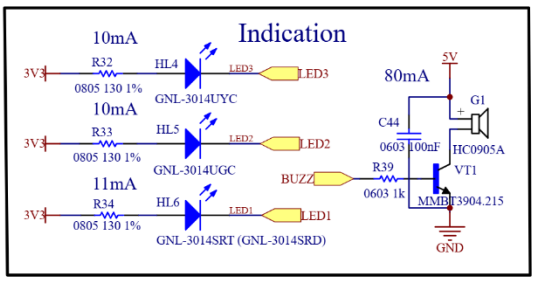

RS-485
GND
3V3
3V3
R10
0603 10k
GND
C10
0603 100nF
R20
0603 560
RS485_P
D6
R21
0603 120
USART2_Rx
RS485_Cntr
USART2_Tx
RO
RE
DE
DI
VCC
B
A
GND
ST3485
X6
Data+(B)
Data-(A)
DS1021_1x2
GND
RS485_N
R11
0603 10k
R22
0603 560
GND
3V3

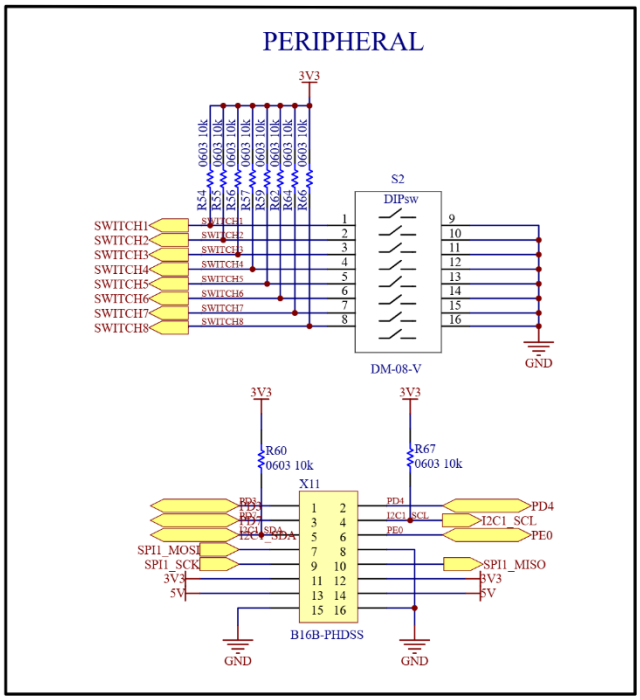

EEPROM
3V3
3V3
R1
0603 100k
5mA
C3
GND
0603 100nF
D2
EEP_CS
SPI1_MISO
nW
S
Q
W
VSS
VCC
HOLD
C
D
nHOLD
SCK
MOSI
nHOLD
SPI1_SCK
SPI1_MOSI
M95256(2M)
R5
0603 100k
GND
GND


MicroSD
3V3
0603 10k
0603 10k
0603 10k
0603 10k
0603 10k
0603 10k
20mA
3V3
R12
R13
R14
R15
R17
R18
X4
C11
0603 100nF
GND
SDIO_D0
SDIO_D1
SDIO_D2
SDIO_D3
SDIO_CMD
SDIO_CLK
D0
D1
D2
D3
CMD
CLK
3V3
GND
SHLD
SHLD
SHLD
microSD_500901-0801
SHLD


ETHERNET RMII
150+25+2LEDS [mA]
ETH_PHY
DP83848IVV (DP83848CVVNOPB)
TRJ0011TBNL (J0011D018)
SP0504BAHT
RJ45
BQ3
CFPS-73
D10
VD8
X12


DEBUG + SW/JTAG
DS1021-2*4SF11-B
DS1021-2*3SF11-B
DS1021_1x4
X9
X8
X10


Indication
GNL-3014UYC
GNL-3014UGC
GNL-3014SRT (GNL-3014SRD)
HC0905A
MMBT3904.215


PERIPHERAL
DM-08-V
B16B-PHDSS
S2
X11

# Component List:

| No. | Component | Designation |
|---|---|---|
| 1. | MCU STM32F407VET6 (LQFP100) | D8 |
| 2. | Ethernet PHY DP83848IVV | D10 |
| 3. | CAN Transceiver SN65HVD230QDR | D3 |
| 4. | USB-to-UART Bridge FT230XS-U | D4 |
| 5. | RS-485 Transceiver ST3485 | D6 |
| 6. | RS-232 Transceiver ADM3202ARNZ | D9 |
| 7. | SPI Flash AT45DB161E | D1 |
| 8. | EEPROM M95256 | D2 |
| | | |
| 10. | LDO Regulator LM1117IMPX-3.3/NOPB | D7 |
| 11. | NPN Transistor MMBT3904 | VT1 |
| 12. | Crystals / Oscillator (32.768kHz FC-135, 8MHz HC49SM, CFPS-73) | BQ1, BQ2, BQ3 |
| 13. | TVS / ESD Protection (SM15T39CA, SP0503BAHTG, SP0504BAHT) | VD1, VD2, VD5, VD8 |
| 14. | Diodes (VS-30BQ040, MBR130LSFT1G, SS34, BAT54C) | VD3, VD4, VD6, VD7 |
| 15. | Resettable Fuses (MF-SM075/60, MF-MSMF050-2) | F1, F2 |
| 16. | Inductors (SRR1208-680YL, CM453232-101KL) | L1, L2 |
| 17. | Buzzer HC0905A | G1 |
| 18. | LED | HL1–HL6 |
| 19. | Resistor | R1–R70 |
| 20. | Capacitor (ceramic, tantalum, electrolytic) | C1–C54 |
| 21. | USB Mini-B Connector | X1 |
| 22. | Power Input Terminal (2-pin) | X2 |
| 23. | microSD Socket | X4 |
| 24. | DB9 Connector (DRB-9FA) | X5, X7 |
| 25. | RJ45 with Magnetics TRJ0011TBNL | X12 |
| 26. | Header Pins (1x2, 2x3, 2x4, 1x4, 16-pin) | X3, X6, X8, X9, X10 |
| 27. | Tactile Switch / 8-position DIP Switch | S1, S2 |
| 28. | Battery Holder BK-912-TR | J1 |
| 29 | JST B16B-PHDSS 2×8 connector | X11 |

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## Datasheets:

- STM32F407VET6.pdf
- DP83848IVV.pdf
- SN65HVD230QDR.pdf
- FT230XS-U.pdf
- ST3485.pdf
- ADM3202ARNZ.pdf
- LM2596S-5.0.pdf
- LM1117-3.3.pdf
- AT45DB161E.pdf
- M95256.pdf
- TRJ0011TBNL.pdf

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## Note:

If your designing software doesn't have any footprint of any particular component, you may make or download a suitable footprint from the internet. In that case, follow the dimensions given in the datasheets.